

CO2xN: Porewater Ammonium

May 2026 Samples

2026-07-23

Contents

0.1	Assessing standard Curves	4
0.2	Update Standard Log	6
0.3	Dilution Corrections - ensure the latest dilution is kept	7
0.4	Performance Check	7
0.5	Analyze the Check Standards	8
0.6	Analyze Blanks	9
0.7	Analyze Duplicates	10
0.8	Spikes	11
0.9	Matrix Effects	12
0.10	Create Log for QAQC	13
0.11	Merge samples with Metadata & check all samples are present	16
0.12	Organize Data	17
0.13	Visualize Data	17
0.14	Export Processed Data	18

##Run Information

```
cat("Run Information: Brenden Blakley") #lets you know what section you're in
```

Run Information: Brenden Blakley

```
library(here)
```

here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW

```
#set the run date & user name
```

```
sample_year <- "2026"
```

```
sample_month <- "May"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
NH3_files <- c("Raw Data/20260608_2M_Phragrr_May_Nh4.csv",  
              "Raw Data/20260626_2M_SMARTXrr_May2026.csv",  
              "Raw Data/20260608_2m_GENXrr_May_NH4.csv",  
              "Raw Data/20260624_GENXrr_2Mrr_May2026.csv"  
            )
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
file_path <- here("Porewater QAQC Logs", "GCREW_NH4_QAQC_Log.csv")
```

```
file_path_QAQC <- here("Porewater QAQC Logs", "GCREW_SEAL_Check_QAQC_Log.csv")
```

```
final_path <- "Processed Data/CO2xN_Porewater_NH4_202605.csv"
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- ""
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

##Setup

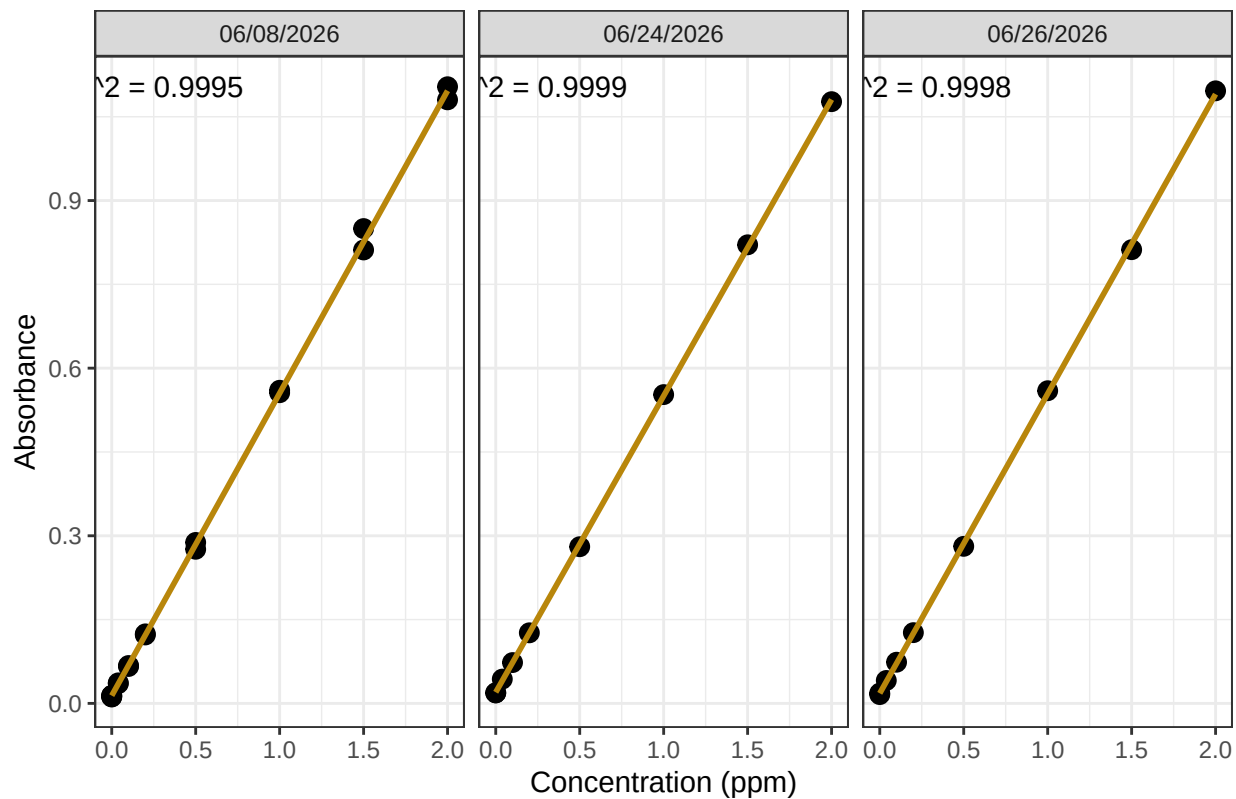
##Read in metadata and create similar sample IDs for matching to samples

```
##Import Data & Clean
```

0.1 Assessing standard Curves

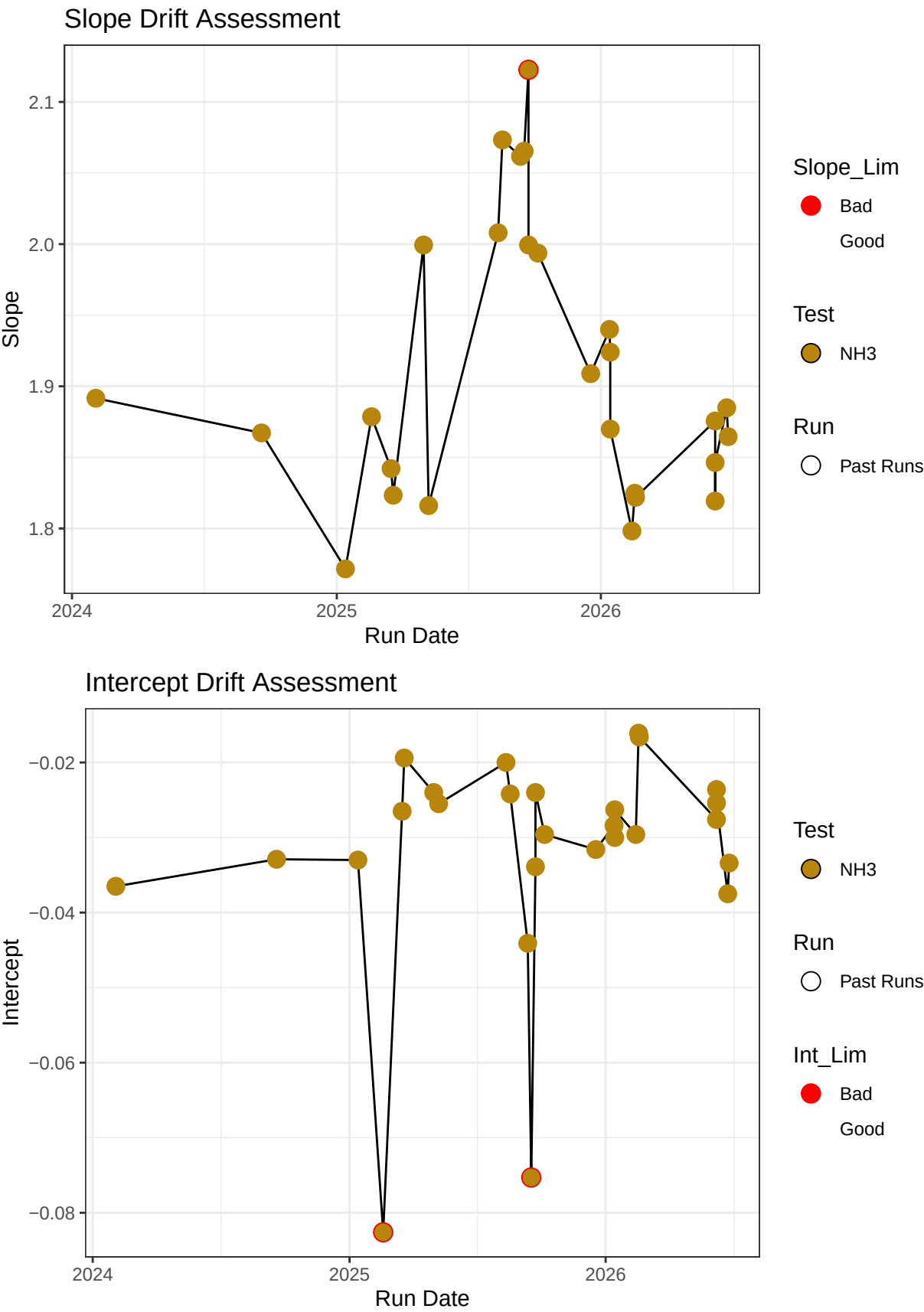
'geom_smooth()' using formula = 'y ~ x'

NH3 Standard Curve



Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	06/08/2026	R2_Pass	0.9994902	1.846406
Ammonia 2	06/24/2026	R2_Pass	0.9999260	1.884907
Ammonia 2	06/26/2026	R2_Pass	0.9998098	1.864565

0.2 Update Standard Log



0.3 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "Reran Samples: c(\"L18-80\", \"L20-40\", \"L20-80\", \"322-80\", \"Ch4-80\", \"Ch5-80\", \"Ch6-80\")"
```

```
## [1] "Diluted Samples: c(\"L18-80\", \"L20-40\", \"L20-80\", \"322-80\", \"Ch4-80\", \"Ch5-80\", \"Ch6-80\")"
```

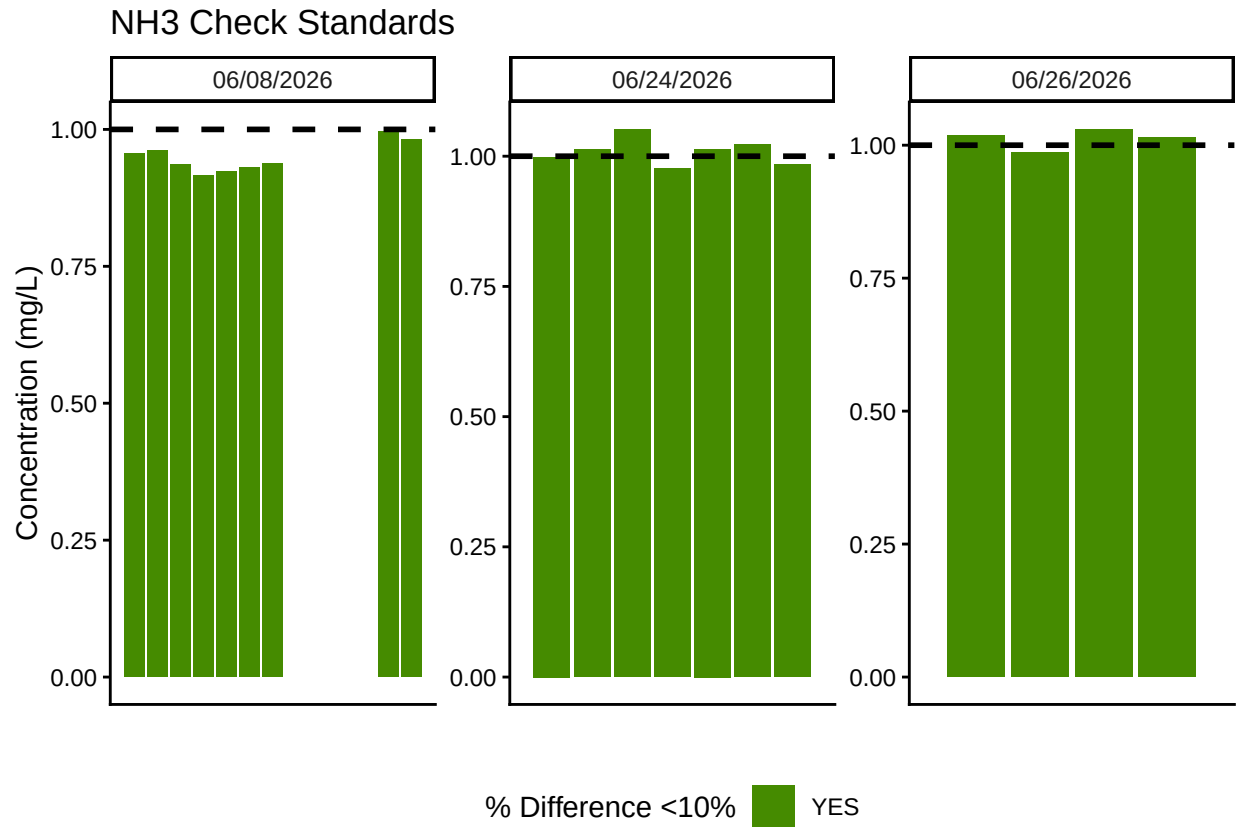
```
## [1] "No Naming Issues Detected"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	06/08/2026	Performance Check Within 25% - PROCEED	1.098202	1.034
Ammonia 2	06/24/2026	Performance Check Within 25% - PROCEED	1.148001	1.034
Ammonia 2	06/26/2026	Performance Check Within 25% - PROCEED	1.046666	1.034

0.5 Analyze the Check Standards

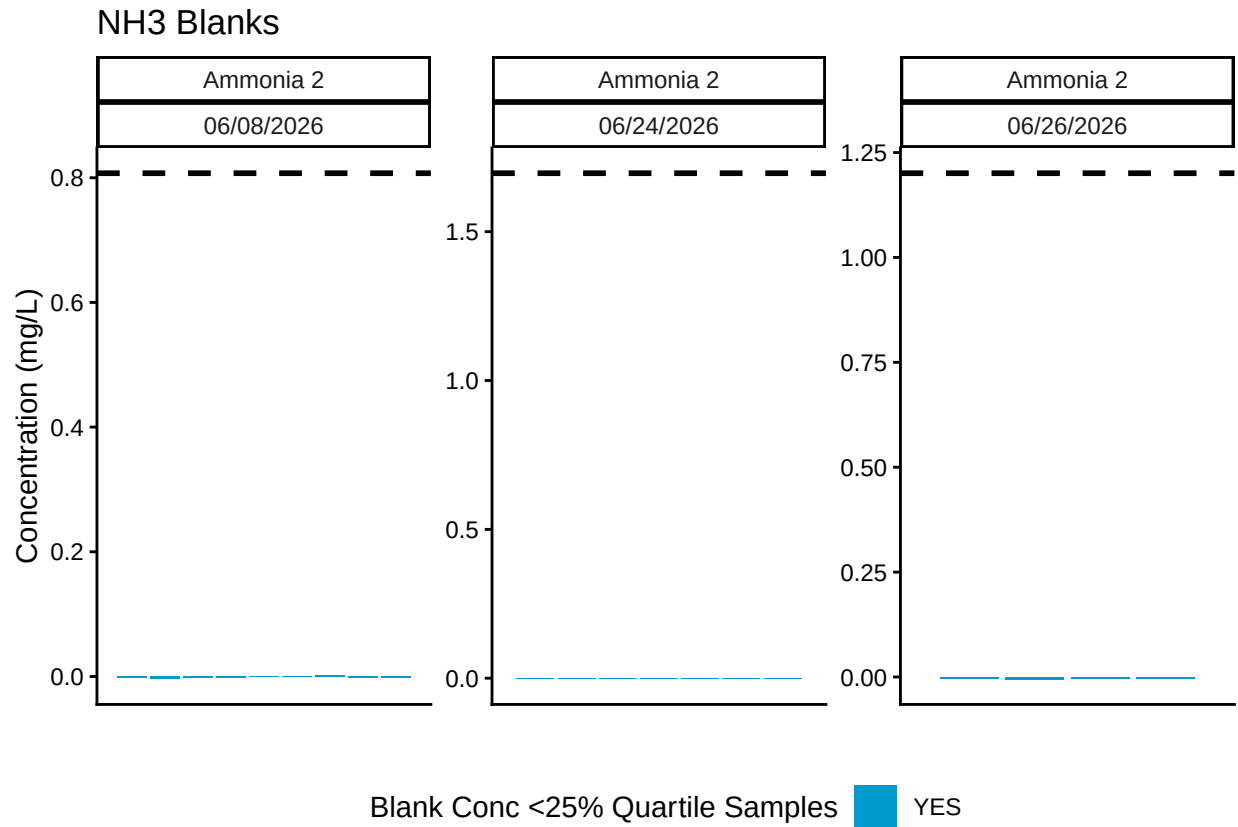
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	06/08/2026	RSV WITHIN RANGE - PROCEED	0.0399642	0.25
Ammonia 2	06/24/2026	RSV WITHIN RANGE - PROCEED	0.0399642	0.25
Ammonia 2	06/26/2026	RSV WITHIN RANGE - PROCEED	0.0399642	0.25



Test	Run_Date	CHK_Flag
Ammonia 2	06/08/2026	>60% of Checks Pass - PROCEED
Ammonia 2	06/24/2026	>60% of Checks Pass - PROCEED
Ammonia 2	06/26/2026	>60% of Checks Pass - PROCEED

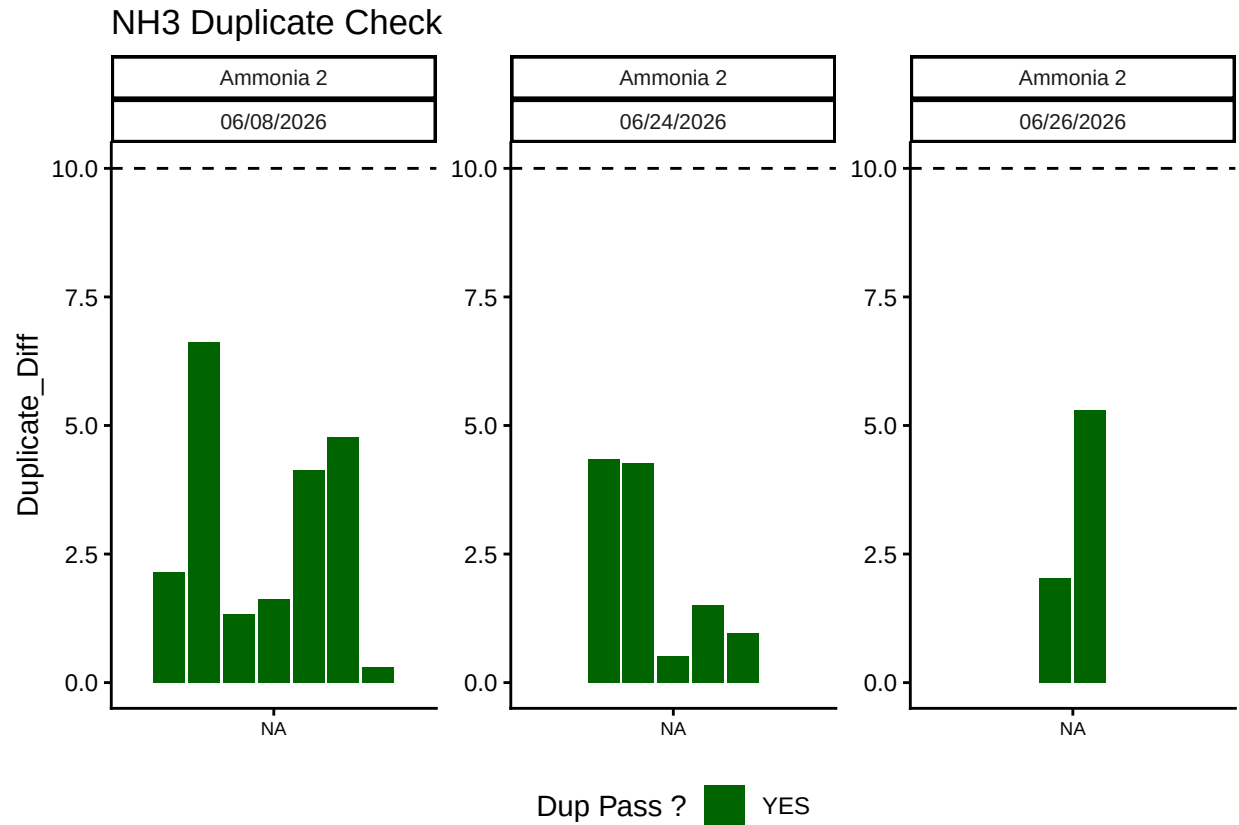
0.6 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blkc_Conc	Quantile_25
Ammonia 2	06/08/2026	>60% of Blanks Pass - PROCEED	-0.0012396	0.807266
Ammonia 2	06/24/2026	>60% of Blanks Pass - PROCEED	-0.0005560	1.696431
Ammonia 2	06/26/2026	>60% of Blanks Pass - PROCEED	-0.0043263	1.200766



0.7 Analyze Duplicates

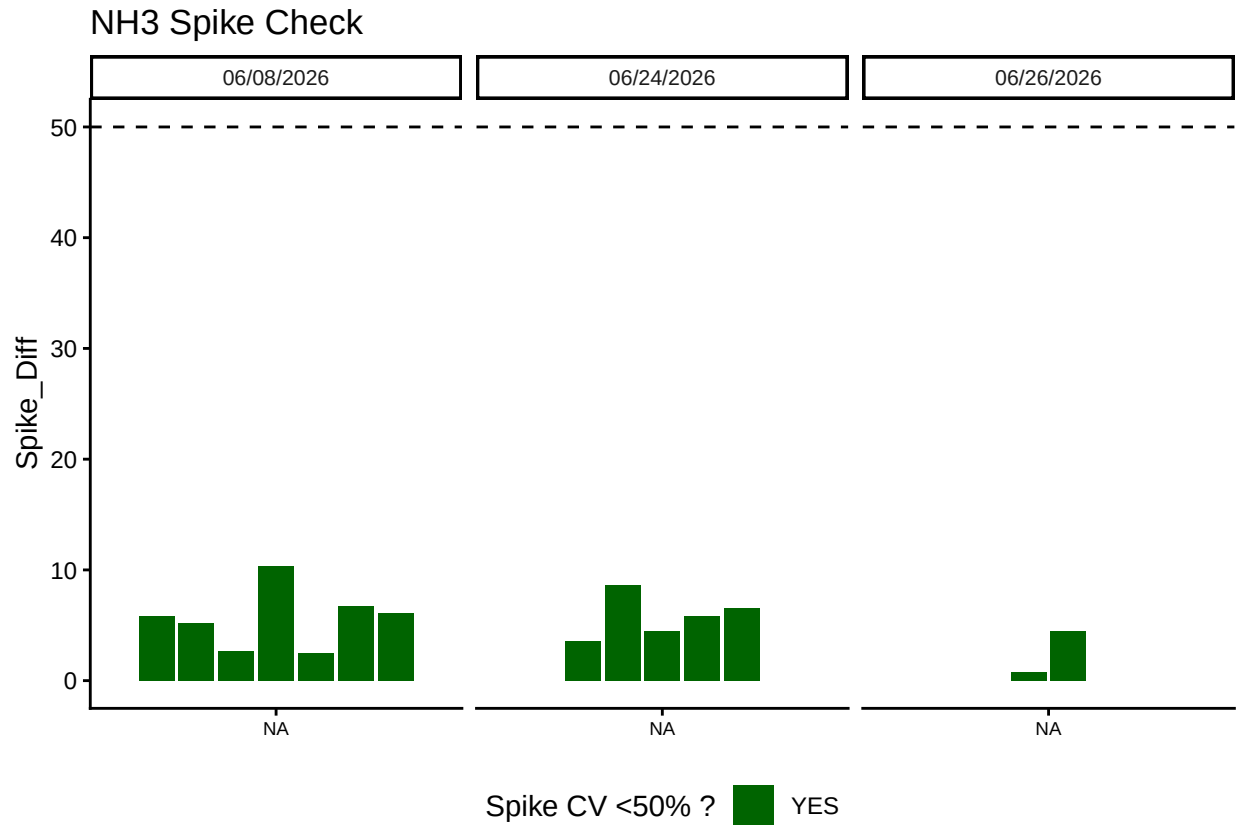
```
## Warning in left_join(Duplicates, Dup_ready, by = join_by(Date_Num)): Detected an unexpected many-to-many
## i Row 1 of 'x' matches multiple rows in 'y'.
## i Row 21 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
## "many-to-many"' to silence this warning.
```



Test	Run_Date	Dup_Flags
Ammonia 2	06/08/2026	>60% of Dups Pass - PROCEED
Ammonia 2	06/24/2026	>60% of Dups Pass - PROCEED
Ammonia 2	06/26/2026	>60% of Dups Pass - PROCEED

0.8 Spikes

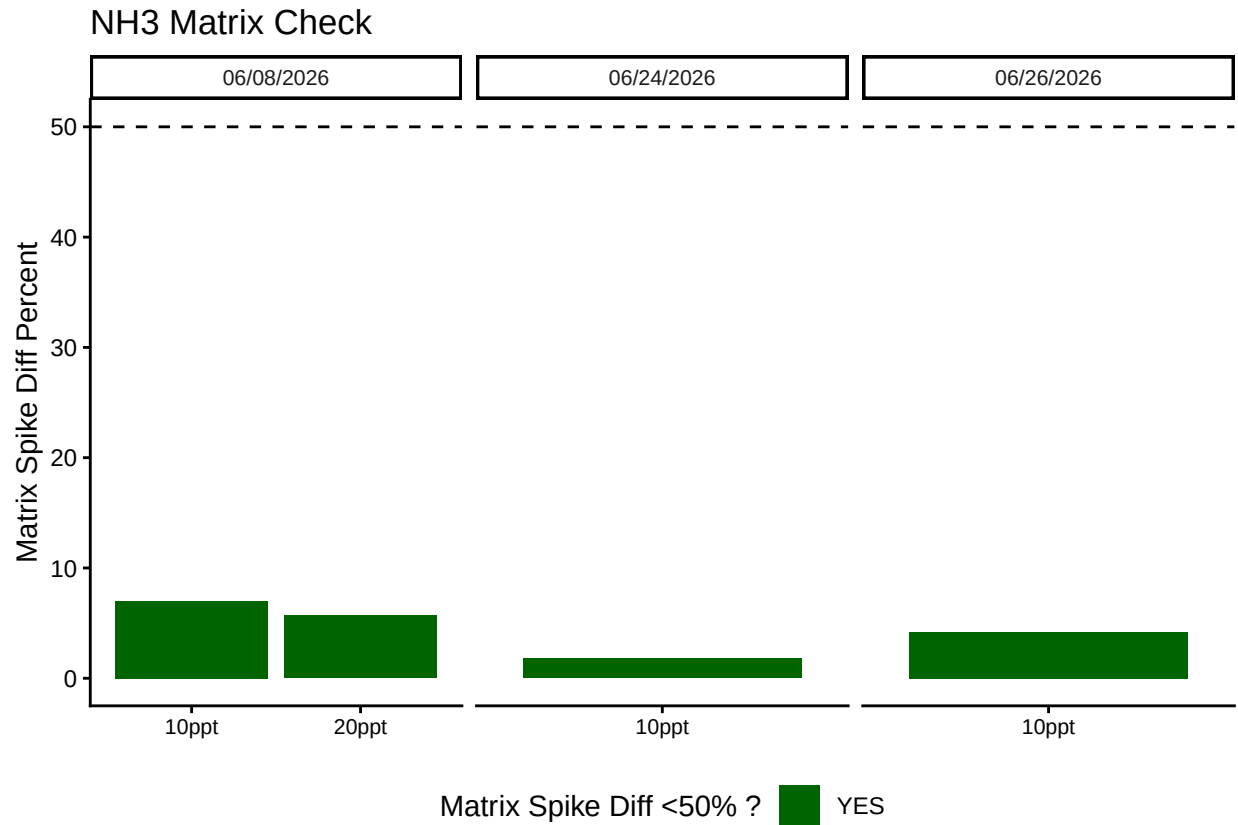
```
## Warning in left_join(Spikes, Spike_ready, by = join_by(Date_Num)): Detected an unexpected many-to-many
## i Row 1 of 'x' matches multiple rows in 'y'.
## i Row 25 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
##   "many-to-many"' to silence this warning.
```



Test	Run_Date	Spike_Flags
Ammonia 2	06/08/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	06/24/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	06/26/2026	>60% of Spikes have a Diff <50% - PROCEED

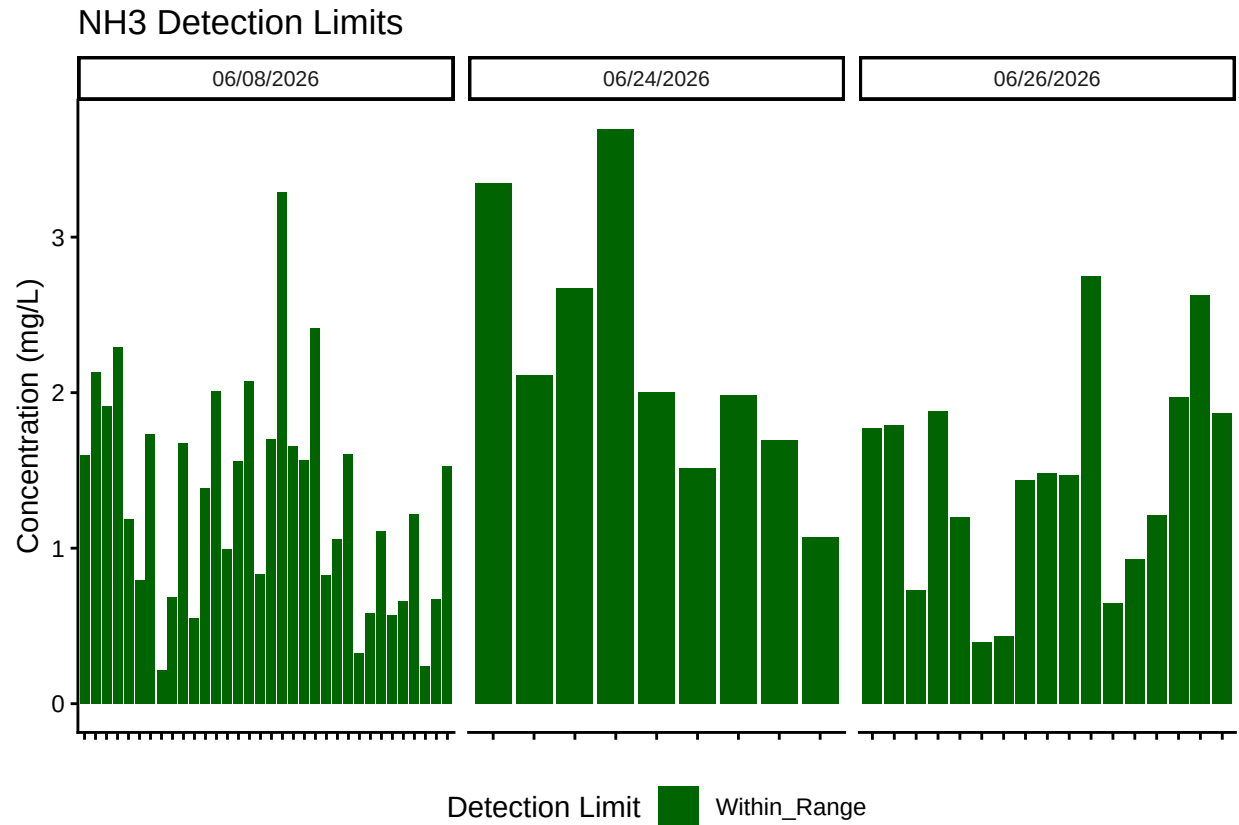
0.9 Matrix Effects

```
## Warning in left_join(Matrix, Spike_ready, by = join_by(Date_Num)): Detected an unexpected many-to-many
## i Row 1 of 'x' matches multiple rows in 'y'.
## i Row 25 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
##   "many-to-many"' to silence this warning.
```



Test	Run_Date	Matrix_Flags
Ammonia 2	06/08/2026	Matrix Has Diff <50% - PROCEED
Ammonia 2	06/24/2026	Matrix Has Diff <50% - PROCEED
Ammonia 2	06/26/2026	Matrix Has Diff <50% - PROCEED

```
##Sample Flagging - Within range of standard curve
```

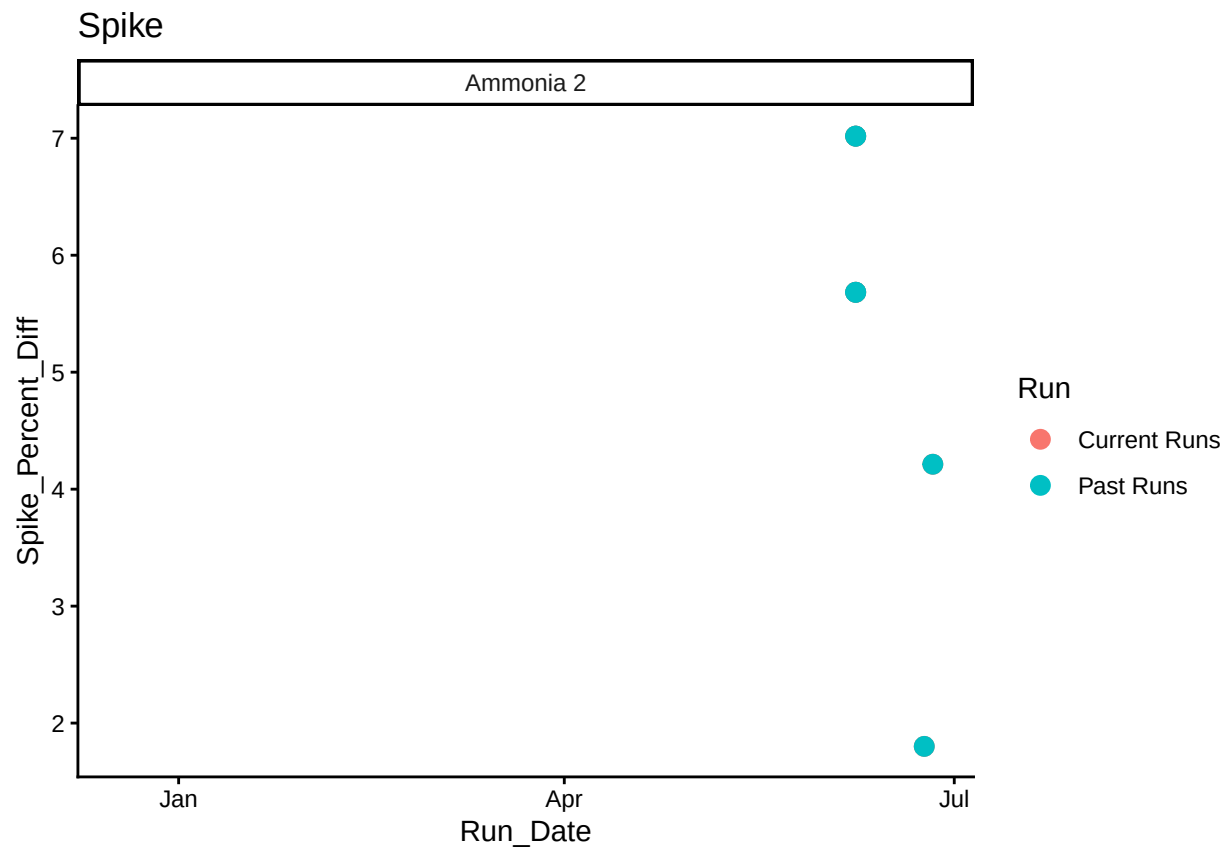


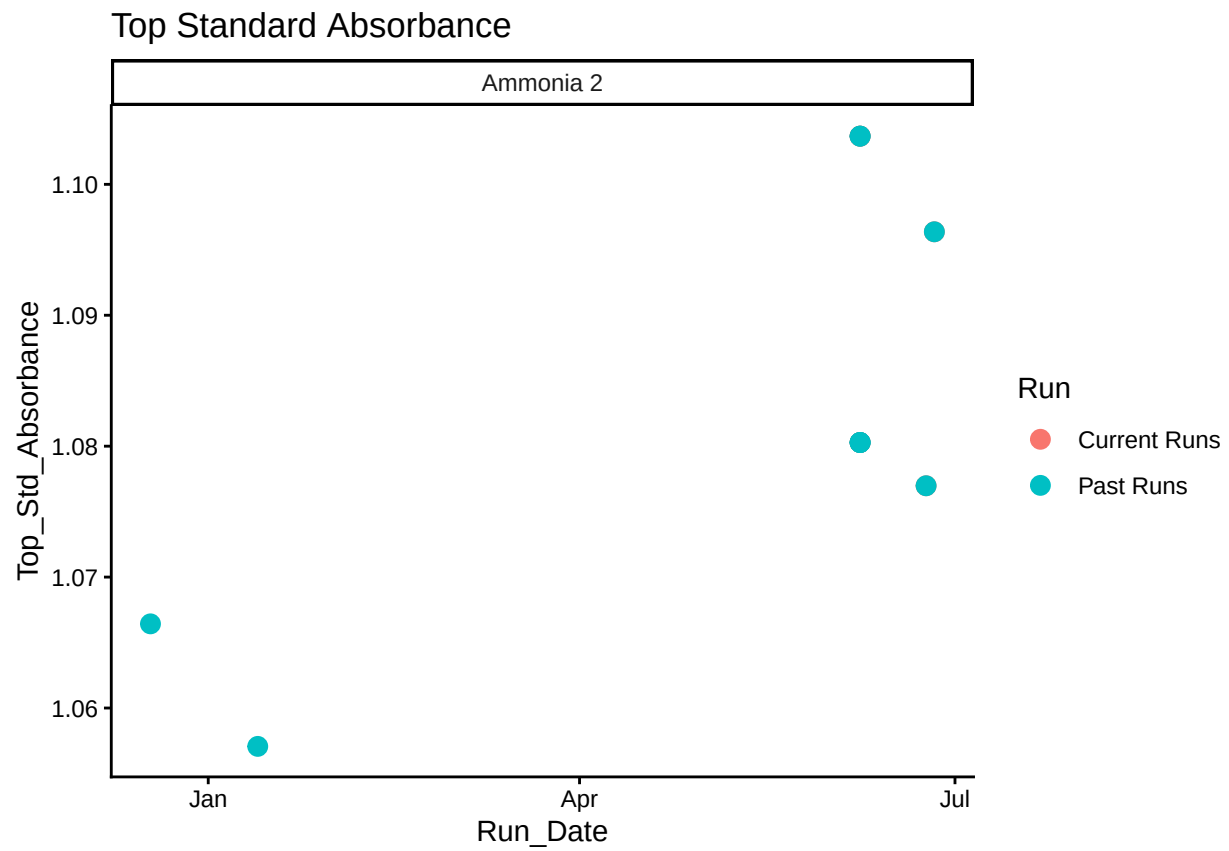
```
##Add QAQC Flags
```

0.10 Create Log for QAQC

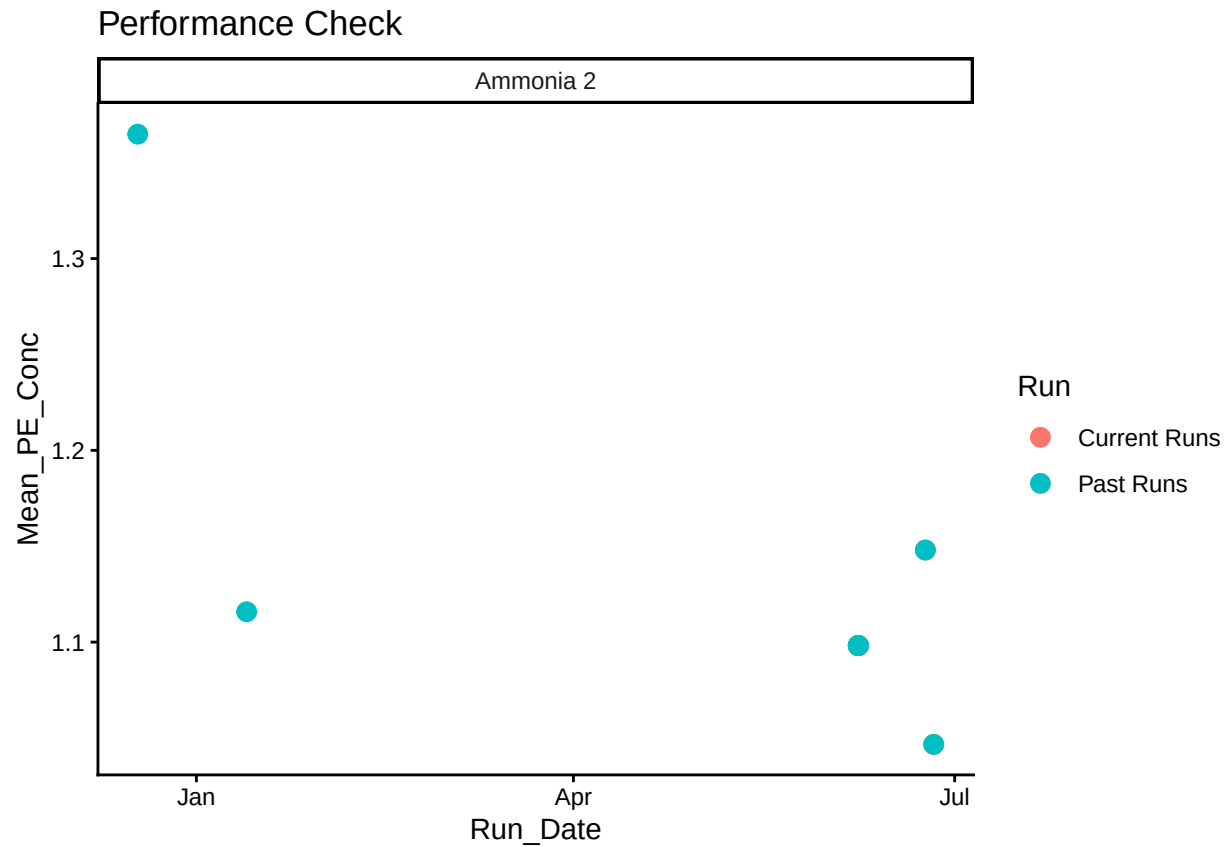
```
## Warning in left_join(., Spikes_Log, by = join_by(Test, Run_Date)): Detected an unexpected many-to-many
## i Row 1 of 'x' matches multiple rows in 'y'.
## i Row 1 of 'y' matches multiple rows in 'x'.
## i If a many-to-many relationship is expected, set 'relationship =
##   "many-to-many"' to silence this warning.

## Warning: Removed 3 rows containing missing values or values outside the scale range
## ('geom_point()').
```





```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_point()').
```



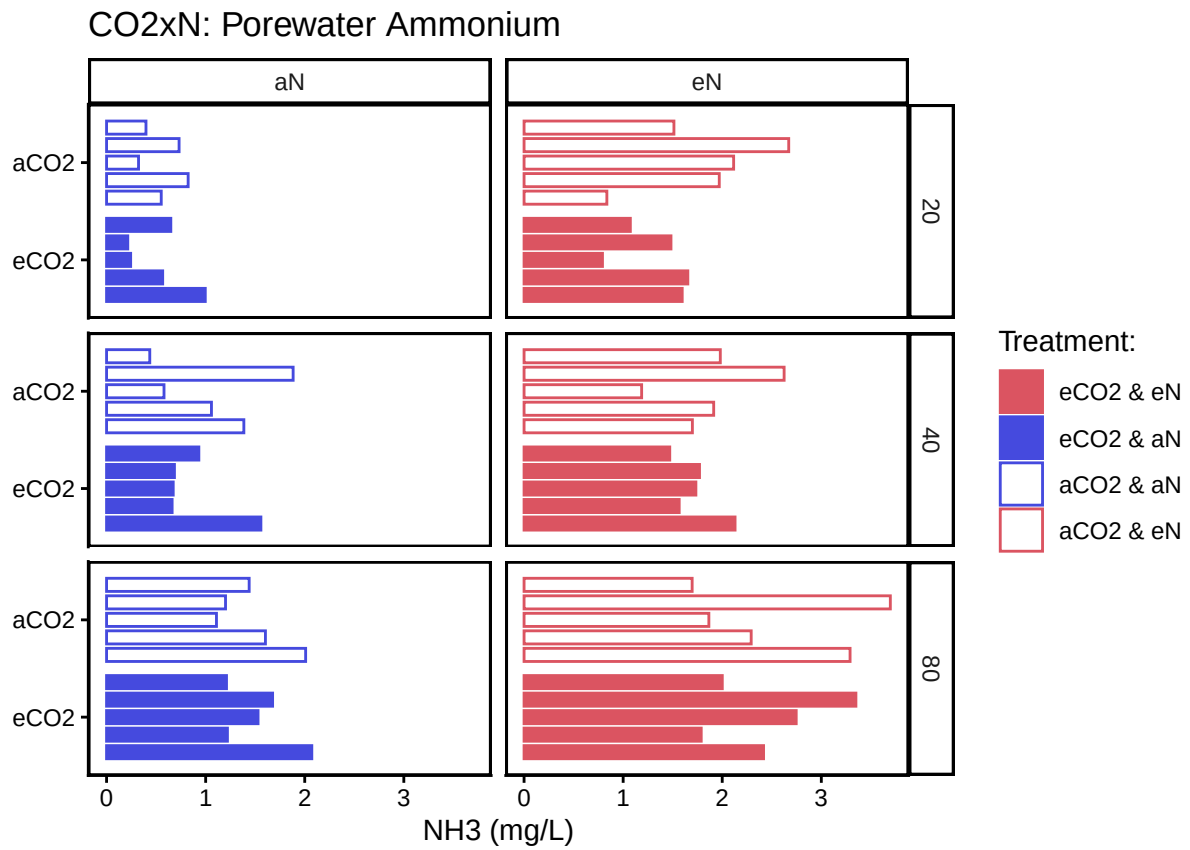
0.11 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

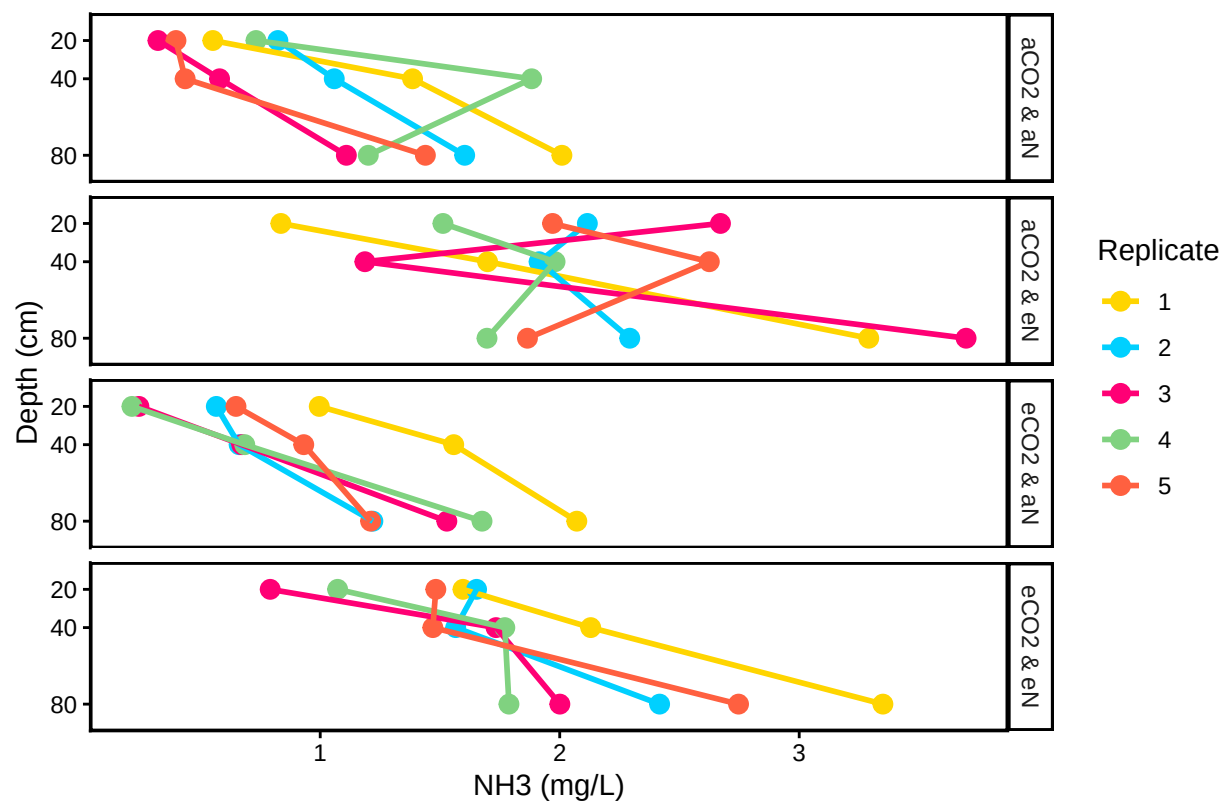
```
## All sample IDs are present in data
```

0.12 Organize Data

0.13 Visualize Data



CO2xN: Porewater NH3 by Replicate



0.14 Export Processed Data

#end